Hands-On Lab (HOL): Create and Connect to a Virtual Machine in AWS

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1. What is a Virtual Machine (VM)?

A **Virtual Machine** is a software-based computer that works like a real physical computer. It has its own operating system, CPU, memory (RAM), storage, and network.

Instead of buying a physical server, we create a VM on top of a physical machine using virtualization technology.

Think of it like this:

- One physical server = one big building
- Virtual machines = many flats inside that building

Each VM works independently.

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2. How Does a Virtual Machine Work?

A **hypervisor** sits on top of the physical hardware.

The hypervisor:

- Divides CPU, RAM, storage, and network
- Gives these resources to multiple virtual machines
- Makes sure VMs are isolated from each other

Each VM:

- Has its own OS (Linux / Windows)
- Can run applications
- Can be started, stopped, or deleted anytime

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3. What is Amazon EC2?

Amazon EC2 (Elastic Compute Cloud) is the AWS service used to create and manage virtual machines in the cloud.

In simple words:

- **EC2 = Virtual Machine in AWS**

With EC2, you can:

- Create Linux or Windows servers
- Choose CPU, RAM, storage
- Start / Stop / Terminate servers anytime
- Pay only for what you use

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4. Types of EC2 Instances in AWS (Simple Explanation)

AWS provides different EC2 types based on workload:

1. General Purpose

Used for normal applications Example: web servers, small apps

2. Compute Optimized

Used for CPU-heavy tasks Example: batch processing, gaming servers

3. Memory Optimized

Used when application needs more RAM Example: databases, in-memory cache

4. Storage Optimized

Used for high disk performance Example: big data, log processing

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5. Hands-On Lab: Create a Virtual Machine (EC2) in AWS

Lab Objective

By the end of this lab, you will:

- Create an EC2 virtual machine
- Configure all components
- Connect to the VM using SSH

Step 1: Login to AWS Console

1. Open browser
2. Go to https://aws.amazon.com
3. Click **Sign In to the Console**
4. Login using your AWS account

Step 2: Open EC2 Service

1. In AWS Console search bar, type **EC2**
2. Click on **EC2**
3. EC2 Dashboard will open

Step 3: Launch a New EC2 Instance

1. Click **Launch Instance**
2. Click **Launch Instance** again

Step 4: Basic Configuration (Name and OS)

4.1 Name the Instance

- Instance Name: `My-First-EC2-VM`

4.2 Choose Amazon Machine Image (AMI)

AMI = Operating System

Choose:

- **Amazon Linux 2** (Free Tier eligible)

Step 5: Choose Instance Type (CPU & RAM)

Before selecting anything, let us understand **what an instance type actually means**.

An **instance type** defines the **hardware capacity** of your virtual machine. It decides:

- How many CPUs your VM will have
- How much RAM (memory) it will get
- How powerful your VM will be

In real life terms:

- Instance type = configuration of a laptop or server

For example:

- Low instance type → low CPU, low RAM (basic work)
- High instance type → more CPU, more RAM (heavy applications)

Why do we need to choose an instance type?

Because different applications need different power:

- Small website → low CPU & RAM
- Database → more RAM
- Data processing → more CPU

Why are we choosing t2.micro here?

- It is **Free Tier eligible**
- Good for learning and practice
- Enough for basic Linux commands and small apps

👉 Choose:

- **t2.micro**

Step 6: Key Pair (Login Security)

Before creating the key pair, understand **why it is needed**.

A **key pair** is used to **securely log in** to your virtual machine.

It works like:

- Username + password (but more secure)

AWS does NOT allow password-based login by default. Instead, it uses:

- **Public key** (stored inside EC2)
- **Private key** (stored with you)

Only the person who has the private key can access the VM.

Why key pair is important?

- Prevents unauthorized access

- More secure than passwords
- Mandatory for Linux EC2 instances

Steps to create key pair

1. Click **Create new key pair**
2. Key pair name: `my-ec2-key`
3. Key pair type: RSA
4. Private key format: `.pem`
5. Click **Create key pair**

👉🏿 **Important**:

- Download the key file immediately
- If you lose it, you cannot log in again

Step 7: Network Settings (Important)

Network settings decide **how your virtual machine communicates** with the outside world.

In simple words:

- Network = how your VM gets internet and access

7.1 VPC (Virtual Private Cloud)

A **VPC** is a private network inside AWS.

Think of it like:

- Your own private office network in the cloud

Why default VPC?

- Already created by AWS
- Easy for beginners
- No manual setup needed

👉 Use **Default VPC**

7.2 Subnet

A **subnet** is a smaller network inside a VPC.

Why subnet is needed?

- Helps in organizing resources
- Decides availability zone

👉 Use **Default subnet**

7.3 Auto-assign Public IP

Public IP allows your VM to be accessed from the internet.

Why enable it?

- Without public IP, you cannot SSH from your laptop

👉 Enable **Auto-assign Public IP**

7.4 Security Group (Firewall)

A **security group** acts like a firewall for your VM.

It decides:

- Which traffic is allowed
- Which traffic is blocked

Why do we need it?

- To protect the VM from attacks
- To allow only required ports

Create new security group:

- Name: `ec2-ssh-sg`

Allow rule:

- SSH | Port 22 | Source: My IP

This means:

- Only your system can connect using SSH

Step 8: Configure Storage

Storage is the **hard disk** of your virtual machine.

This is where:

- OS files are stored
- Applications are installed
- Logs and data are saved

Why storage is required?

Without storage:

- OS cannot boot
- Data cannot be saved

Storage settings explanation

1. **Root volume size**:

2. Default: 8 GB

3. Enough for learning and basic usage

4. **Volume type**:

5. gp2 / gp3 (General Purpose SSD)

6. Balanced performance and cost

👉 Keep default values for this lab

Step 9: Review and Launch

1. Review all settings
2. Click **Launch Instance**
3. Instance will start in few seconds

Step 10: Check Instance Status

1. Go to **EC2 → Instances**
2. Instance state should be **Running**
3. Status check should be **2/2 checks passed**

Step 11: Connect to EC2 (Linux VM)

11.1 Copy Public IP

From EC2 instance details:

- Copy **Public IPv4 address**

11.2 Connect Using SSH (Linux / Mac)

1. Open terminal
2. Go to folder where key is saved

Run:

```
chmod 400 my-ec2-key.pem
ssh -i my-ec2-key.pem ec2-user@<Public-IP>
```

11.3 Connect Using SSH (Windows – PuTTY)

1. Convert `.pem` to `.ppk` using PuTTYgen
2. Open PuTTY
3. Enter Public IP
4. Use private key in SSH Auth
5. Login as user: `ec2-user`

Step 12: Verify VM is Working

After login, run:

```
uname -a
whoami
```

You are now inside your AWS Virtual Machine 🦻

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6. Clean Up (Very Important)

To avoid billing:

1. Go to EC2 → Instances
2. Select your instance
3. Click **Instance state → Terminate instance**

Lab Summary

In this lab, you learned:

- What a virtual machine is
- How EC2 works in AWS
- How to configure instance type, storage, network
- How to connect to EC2 using SSH

👈🏼 End of Hands-On Lab